Computer Networks Lab

Cisco Packet Tracer — Assignment 4

All practicals must be completed inside **Cisco Packet Tracer**.
Each assignment requires exactly **two screenshots** as submission.

Tool: Cisco Packet Tracer

How to Submit

- Complete the practical entirely inside **Cisco Packet Tracer**. No other tool is accepted.
- Save your file as `RollNo_Set4.pkt` (e.g. `20221234_Set4.pkt`) and submit it.
- Along with the file, submit exactly **two screenshots** as described at the end of each set.
- Screenshots must be clear and show all device labels and relevant details.

1

Scenario

DevNest Technologies has two departments, **Development** and **Testing**, currently connected to the same switch and placed on the same network. This is causing unnecessary broadcast traffic and makes it difficult to keep the two teams' data separate.

The network administrator, Ms. Laleh Ahmadi, wants the two departments split into **separate logical networks**, each with its own addressing, while still being able to reach each other when required. However, the company has decided **not to purchase a separate router** — she insists the existing switch infrastructure alone should be capable of handling both the separation and the communication between the two networks.

Device	IP Address	Default Gateway
Dev-PC1	192.168.10.10	192.168.10.1
Dev-PC2	192.168.10.20	192.168.10.1
Test-PC1	192.168.20.10	192.168.20.1
Test-PC2	192.168.20.20	192.168.20.1
Subnet Mask: `255.255.255.0` on all PCs		

Submission

Submit the following **two screenshots**:

1. The completed **topology** in Packet Tracer Logical View showing the switch and all six PCs connected, with the two groups clearly distinguishable and IP labels visible on every PC.
2. A **ping result** in the Command Prompt of Dev-PC1 successfully reaching Test-PC1 across the two networks, showing zero packet loss.

Note: Not every switch is capable of this. You will need to pick a switch model from the device list that supports routing between networks, not a standard access-layer switch. There is also a single global setting that must be turned on for this capability to actually work — without it, everything may look correctly configured, yet the two groups still won't be able to reach each other.